

Research Interests	Computer Systems, Data Systems, Distributed Systems, Systems for Machine Learning	
Education	Massachusetts Institute of Technology	2020 – Present
	Doctor of Philosophy Student in Computer Science (PhD)	Cambridge, MA
	Advisor: Tim Kraska, Data Systems Group	
	University of Toronto	2018 – 2020
	Master of Science in Computer Science (MSc)	Toronto, ON
	- Advisor: Gennady Pekhimenko, EcoSystem Group (Computer Systems and Networks) - Thesis: Habitat: Prediction-guided Hardware Selection for Deep Neural Network Training	
	University of Waterloo	2013 – 2018
	Bachelor of Software Engineering (BSE)	Waterloo, ON
	94.65% Cumulative Average, Graduated with Distinction and on the Dean’s Honors List	
Publications	TreeLine: An Update-In-Place Key-Value Store for Modern Storage	
	Geoffrey X. Yu *, Markos Markakis*, Andreas Kipf*, Per-Åke Larson, Umar Farooq Minhas, and Tim Kraska	
	Proceedings of the VLDB Endowment (VLDB), 2023 (<i>to appear</i>)	
	Habitat: A Runtime-Based Computational Performance Predictor for Deep Neural Network Training	
	Geoffrey X. Yu , Yubo Gao, Pavel Golikov, and Gennady Pekhimenko	
	USENIX Annual Technical Conference (USENIX ATC), 2021	
	Skyline: Interactive In-Editor Computational Performance Profiling for Deep Neural Network Training	
	Geoffrey X. Yu , Tovi Grossman, and Gennady Pekhimenko	
	ACM Symposium on User Interface Software and Technology (UIST), 2020	
* Denotes equal contribution.		
Demonstrations	Skyline: Interactive In-Editor Performance Visualizations and Debugging for DNN Training	
	Geoffrey X. Yu , Tovi Grossman, and Gennady Pekhimenko	
	Conference on Machine Learning and Systems (MLSys), 2020	
	TBD Suite: Benchmarking and Profiling Tools for DNNs	
	Geoffrey X. Yu , Hongyu Zhu, Anand Jayarajan, Bojian Zheng, Abhishek Tiwari, and Gennady Pekhimenko	
	Conference on Machine Learning and Systems (MLSys), 2019	
Teaching Experience	Teaching Assistant	Massachusetts Institute of Technology
	6.5830 / 6.5831 – Database Systems	Fall 2022
	Teaching Assistant	University of Toronto
	CSC 209 – Software Tools and Systems Programming	Fall 2018, Winter 2019
Professional Experience	Intel Corporation	Research Graduate Intern
	North America – Remote	May – August 2021
	- Mentors: Nesime Tatbul and David Cohen - Led research on developing new designs for concurrent learned indexes that support database transactions.	

Facebook
Menlo Park, CA

Software Engineering Intern
September – December 2017

- Utilized C++ to develop a new queuing service for the Async Tier—Facebook’s asynchronous job execution platform, responsible for handling tens of millions of jobs per minute.
- Designed and implemented a mechanism to reschedule failed jobs, requiring development across new and legacy infrastructure services on the Async Tier.
- Improved operational efficiency by implementing scripts to facilitate database shard management.

Facebook
Menlo Park, CA

Software Engineering Intern
January – April 2017

- Implemented robust Java code diagnostics for [Nuclide](#), enabling Java editing support for massive projects with over 60,000 source files.
- Engineered and developed Nuclide’s Java type resolution mechanism, which was capable of handling source code files with more than 10,000 transitive dependencies.
- Added partial rendering to the Nuclide console, increasing the number of entries that could be retained in the user interface by 20 times.

Apple
Cupertino, CA

Software Engineering Intern
May – August 2016

- [Contributed to Apache Cassandra](#) to improve disaster recovery across data centers.
- Prototyped a Cassandra partition blacklisting mechanism in Java to support nodes in production.
- Improved Cassandra timeout enforcement and introduced custom per-query timeouts
- Enhanced Cassandra error handling to enable error propagation from replica nodes to clients.

Coursera
Mountain View, CA

Software Engineering Intern
September – December 2015

- Developed a near real-time course tracking tool by leveraging Scala, Kafka, and JavaScript with React—providing an aggregate view of over 1700 courses on Coursera.
- Wrote a scheduled job in Scala to validate user role invariants across multiple data stores.
- Utilized Scala and JavaScript with React to build a suite of administrative course editing tools end-to-end, replacing legacy tools that lacked comprehensive validations.

Coursera
Mountain View, CA

Software Engineering Intern
January – April 2015

- Engineered robust course authoring tools using React and Flux that were made available to over 100 Coursera partner institutions.
- Built the front end for an instructor to learner mass email communication tool. This tool has been used to send out over 800,000 emails.

Milq (now Layer 6, acquired by TD Bank Group)
Toronto, ON

Full Stack Software Engineering Intern
May – August 2014

- Developed the tag page—an aggregation of content about a specific hashtag.
- Engineered scalable data API endpoints in Node.js using Redis and MongoDB to deliver matched content to the tag page.
- Developed functionality that fetched, resized, and stored video thumbnails into an Amazon S3 bucket for new and existing content added to Milq.

Awards and Honors

- **NSERC Postgraduate Scholarship – Doctoral (PGS D)** 2020 – 2023
Merit-based doctoral research scholarship awarded by the Government of Canada (63,000 CAD).
- **NSERC Alexander Graham Bell Canada Graduate Scholarship (CGS D)** 2020 – 2023 (declined)
Merit-based doctoral research scholarship awarded by the Government of Canada (105,000 CAD).

- **NSERC Canada Graduate Scholarship – Master’s (CGS M)**
Merit-based master’s research scholarship awarded by the Government of Canada (17,500 CAD).

2019 – 2020
- **Snap Research Scholarship**
Awarded by Snap Inc. for research quality, technical skills, and communication skills (10,000 USD).

January 2019
- **Vector Institute Scholarship in Artificial Intelligence**
Awarded to top students pursuing AI-related master’s studies in Ontario (17,500 CAD).

2018 – 2019
- **Queen Elizabeth II Graduate Scholarship in Science and Technology**
Merit-based scholarship awarded to students pursuing graduate studies in Ontario (15,000 CAD).

2018 – 2019
- **Wolfond Fellowship, University of Toronto**
Awarded at the University of Toronto for academic achievements (10,000 CAD).

2018 – 2019
- **Sandford Fleming Foundation Award for Academic Excellence**
Best graduating academic record in the Software Engineering program at the University of Waterloo.

June 2018
- **University of Waterloo Faculty of Engineering First in Class Award**
Highest term average in the Software Engineering program at the University of Waterloo (Fall 2016).

May 2017
- **Wish Scholarship, University of Waterloo**
Awarded by the University of Waterloo based on academic merit.

January 2016
- **University of Waterloo Faculty of Engineering Upper Year Scholarship**
Awarded by the Faculty of Engineering at the University of Waterloo based on academic merit.

May 2015
- **Software Engineering Entrance Scholarship, University of Waterloo**
Awarded to first year Software Engineering students based on academic and extracurricular achievements.

September 2013
- **University of Waterloo President’s Scholarship of Distinction**
Awarded to undergraduate students with an admissions average of 95% or greater.

September 2013
- **Toronto District School Board Top Scholar**
Highest high school graduating average in the Toronto District School Board (Toronto, ON).

June 2013